

BFS and DFS

Aim: To implement Breadth First Search (BFS) and Depth First Search (DFS) for graph traversal using Python.

BFS: BFS visits vertices level by level starting from a given vertex. It uses a queue to keep track of the vertices to be visited.

DFS: DFS explores a path as deeply as possible before backtracking. It uses recursion or a stack for traversal.

Test Cases: The algorithms are tested using different starting vertices such as A, B, and C.

Time Complexity: Both BFS and DFS have a time complexity of $O(V + E)$, where V is the number of vertices and E is the number of edges.

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

# Breadth First Search (BFS)
def bfs(graph, start):
    visited = set()
    queue = deque([start])
    result = []

    while queue:
        vertex = queue.popleft()

        if vertex not in visited:
            visited.add(vertex)
            result.append(vertex)

            for neighbour in graph[vertex]:
                if neighbour not in visited:
                    queue.append(neighbour)

    return result

# Depth First Search (DFS)
def dfs(graph, start):
    visited = set()
```

```
result = []

def dfs_recursive(vertex):
    if vertex in visited:
        return

    visited.add(vertex)
    result.append(vertex)

    for neighbour in graph[vertex]:
        dfs_recursive(neighbour)

dfs_recursive(start)
return result

print("Test Case 1:")
print("Starting vertex: A")
print("BFS:", bfs(graph, 'A'))
print("DFS:", dfs(graph, 'A'))

print("\nTest Case 2:")
print("Starting vertex: B")
print("BFS:", bfs(graph, 'B'))
print("DFS:", dfs(graph, 'B'))

print("\nTest Case 3:")
print("Starting vertex: C")
print("BFS:", bfs(graph, 'C'))
print("DFS:", dfs(graph, 'C'))
```

Test Case 1:
Starting vertex: A
BFS: ['A', 'B', 'C', 'D', 'E', 'F']
DFS: ['A', 'B', 'D', 'E', 'F', 'C']

Test Case 2:
Starting vertex: B
BFS: ['B', 'D', 'E', 'F']
DFS: ['B', 'D', 'E', 'F']

Test Case 3:
Starting vertex: C
BFS: ['C', 'F']
DFS: ['C', 'F']

